

XINGREN, China

WMO: 579020

Lat: 25.4308N

Lon: 105.1832E

Elev: 1379

StdP: 85.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	0.5	-4.2	3.1	7.6	-2.5	3.6	5.9	8.8	18.1	7.3	17.7	1.6	40	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.4	29.7	19.3	28.7	19.7	27.7	19.8	22.4	26.4	22.0	26.0	21.6	25.4	4.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.3	18.9	24.0	20.8	18.4	23.7	20.5	18.0	23.3	74.0	26.8	72.1	26.2	70.5	25.4	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	6.0	5.1	-1.8	32.2	1.1	1.3	-2.6	33.1	-3.2	33.8	-3.9	34.5	-4.7	35.4		
			-2.4	23.4	1.2	0.5	-3.3	23.7	-4.1	24.0	-4.7	24.3	-5.6	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.3	7.2	9.8	13.6	18.1	20.5	21.9	22.9	22.5	20.4	17.0	13.2	8.4
	DBStd	6.51	4.46	5.58	4.81	4.25	3.14	2.30	1.54	1.69	2.49	3.11	3.58	3.45
	HDD10.0	286	112	70	19	1	0	0	0	0	0	1	11	73
	HDD18.3	1354	345	241	159	56	16	2	0	1	9	63	156	307
	CDD10.0	2602	25	66	131	245	326	358	400	388	313	218	108	24
	CDD18.3	629	0	3	13	50	83	111	142	130	72	23	3	0
	CDH23.3	2908	1	29	116	351	447	463	588	570	282	53	8	1
	CDH26.7	479	0	2	19	87	96	68	83	91	32	1	0	0
Wind	WSAvg	2.1	2.1	2.5	2.6	2.6	2.3	2.1	2.0	1.7	1.8	1.9	2.0	2.0
Precipitation	PrecAvg	1332	21	25	29	74	173	258	224	213	136	106	51	22
	PrecMax	1889	40	80	103	147	309	476	474	418	362	406	144	84
	PrecMin	732	9	2	4	3	42	82	57	64	36	25	10	2
	PrecStd	226	8	17	21	39	66	104	109	98	70	77	34	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	26.6	29.0	31.2	31.5	30.0	29.8	30.1	29.3	26.4	24.4	21.0
		MCWB	10.9	11.6	13.8	15.6	17.7	21.5	21.6	21.1	20.7	18.9	15.6	12.4
	2%	DB	18.9	23.9	26.5	29.1	29.0	28.5	28.6	28.7	27.5	24.8	22.2	18.1
		MCWB	11.2	11.7	13.6	15.9	18.8	21.2	21.4	21.0	20.0	18.2	15.3	12.1
	5%	DB	16.4	21.5	24.2	27.3	27.4	27.3	27.6	27.7	26.3	23.5	20.4	15.7
		MCWB	10.8	11.8	14.0	16.4	18.8	21.0	21.2	20.9	19.6	17.7	14.7	11.2
	10%	DB	13.7	18.9	21.5	25.3	25.8	26.1	26.5	26.6	25.0	21.9	18.8	13.6
		MCWB	10.2	11.7	14.0	16.6	18.8	20.7	21.0	20.7	19.2	17.3	14.3	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.1	17.0	19.9	21.7	23.0	23.0	22.7	21.8	20.1	17.8	14.0
		MCDB	17.9	21.4	22.5	25.2	26.4	26.9	26.7	27.0	26.0	24.1	21.4	18.1
	2%	WB	12.0	13.9	16.0	19.1	21.1	22.4	22.3	22.1	21.2	19.3	16.6	12.7
		MCDB	16.6	19.4	21.5	24.2	25.7	26.3	26.1	26.4	25.1	22.8	19.9	16.5
	5%	WB	11.2	12.9	15.2	18.3	20.5	21.8	21.9	21.6	20.7	18.7	15.8	11.7
		MCDB	15.2	18.1	20.7	23.4	24.8	25.7	25.7	25.7	24.3	21.9	18.9	15.4
	10%	WB	10.2	11.9	14.4	17.6	19.9	21.4	21.5	21.2	20.2	18.1	14.8	10.5
		MCDB	13.7	17.1	19.7	22.5	23.9	24.8	25.1	25.0	23.5	21.1	18.0	13.3
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.6	8.3	8.5	7.6	6.2	6.4	7.1	7.0	5.9	6.8	6.0
		MCDBR	11.8	13.9	13.6	12.1	10.4	8.2	8.3	9.2	9.1	8.5	9.8	10.4
		MCWBR	5.4	5.0	4.5	3.8	3.5	2.9	2.7	2.9	2.9	3.1	3.9	5.1
	5% WB	MCDBR	10.2	11.7	11.3	10.3	8.1	6.9	6.9	7.6	7.2	6.9	7.7	9.3
		MCWBR	5.1	5.1	4.8	4.0	3.3	2.9	2.7	2.9	2.7	2.9	3.4	4.8
Clear-Sky Solar Irradiance	taub	0.491	0.488	0.620	0.588	0.500	0.495	0.462	0.504	0.508	0.551	0.519	0.528	
	taud	1.855	1.898	1.624	1.746	2.057	2.110	2.207	2.062	2.013	1.824	1.823	1.761	
	Ebn at Noon	738	784	703	742	808	807	833	799	785	717	705	678	
	Edh at Noon	181	186	257	233	170	160	145	168	172	197	185	192	
All-Sky Solar Radiation	RadAvg	2.28	3.30	3.93	4.72	4.76	4.14	4.72	4.57	3.89	2.82	2.89	2.12	
	RadStd	0.51	0.70	0.43	0.57	0.49	0.54	0.34	0.35	0.41	0.40	0.37	0.36	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	+0.38	N/A	N/A	+0.57	N/A	N/A	N/A	N/A	+118	+68
		+0.43	N/A	N/A	+0.60	+0.20	N/A	N/A	-110	+127	+50

Nomenclature: See separate page